

NTD / FAF5 Network Links Data Quality Evaluation

1. Overview and Description of the Dataset

National Transit Database (NTD) – Freight Analysis Framework (FAF) Dataset available at:
(<https://geodata.bts.gov/maps/b33d6a096bb94ca6b5cf5d96bf68a682>)

The Freight Analysis Framework (FAF), produced through a partnership between Bureau of Transportation Statistics (BTS) and Federal Highway Administration (FHWA), integrates data from a variety of sources to create a comprehensive picture of freight movement among states and major metropolitan areas by all modes of transportation. The current version is FAF5.

The Freight Analysis Framework (FAF5) - Network Links dataset was created from 2017 base year data and was published on April 11, 2022 from the Bureau of Transportation Statistics (BTS) and is part of the U.S. Department of Transportation (USDOT)/Bureau of Transportation Statistics (BTS) National Transportation Atlas Database (NTAD). The FAF (Version 5) Network contains 487,384 link features. All link features are topologically connected to permit network pathbuilding and vehicle assignment using a variety of assignment algorithms. The FAF Link and the FAF Node datasets can be used together to create a network. The link features include all roads represented in prior FAF networks, and all roads in the National Highway System (NHS) and the National Highway Freight Network (NHFN) that are currently open to traffic. Other included links provide connections between intersecting routes, and to select intermodal facilities and all U.S. counties. The network consists of over 588,000 miles of equivalent road mileage. The dataset covers the 48 contiguous States plus the District of Columbia, Alaska, and Hawaii.

2. Data Structure, Schema, and Metadata Assessment

The data set contains 487,394 rows and 53 columns. Where each row represents geospatial transport network link. It includes identifiers, route attributes, geography, facility classification, network designations, directional lane/speed/time fields, toll and border fields, and geometry-related length information.

Rows: 487394

Columns: 53

```
C:\Users\tinew\AppData\Local\Temp\ipykernel_48700\4204625015.py:5: DtypeWarning: Columns (11, 26, 33, 37, 38, 45) have mixed types. Specify dtype option on import or set low_memory=False.  
df = pd.read_csv('./data/csv.csv', delimiter=',')
```

OBJECTID	ID	LENGTH	DIR	DATA1	VERSION	Class	Class_Description	Road_Name	Sign_Rte	
0	1	1	0.054083	1	1324805	V2021.05	19.0	Access/Circulator Road	Facility PETERSBURG FERRY TERMINAL RD	NaN
1	2	2	0.114854	0	1324806	V2021.05	19.0	Access/Circulator Road	Facility PETERSBURG FERRY TERMINAL RD	NaN

The schema includes numeric, text, and categorical fields.

The main schema groups are organized as follows:

- **Identifier fields:** OBJECTID , ID , DATA1 .
- **Link and direction fields:** LENGTH , DIR .
- **Classification fields:** Class , Class_Description , F_Class , Facility_Type .
- **Geography fields:** Country , STATE , STFIPS , County_Name , CTFIPS , Urban_Code , FAFZONE .
- **Route fields:** Road_Name , Sign_Rte , Rte_Type , Rte_Number .
- **Network designation fields:** NHS , STRAHNET , NHFN , Truck .
- **Directional operational fields:** AB_Lanes , BA_Lanes , Speed_Limit , AB_FinalSpeed , BA_FinalSpeed , AB_FreeFlowTime , BA_FreeFlowTime .
- **Toll and border fields:** Toll_Type , Toll_Name , TRUCKTOLL , BorderLink , AddedBorderTime .
- **Geometry-related field:** Shape__Length .

	column	dtype	non_null	missing	missing_pct	unique_values
42	BorderLink	float64	168	487226	99.965531	1
43	AddedBorderTime	float64	168	487226	99.965531	1
33	Toll_Link_Name	object	1135	486259	99.767129	659
32	Toll_Link	float64	1243	486151	99.744970	1
41	TRUCKTOLL	float64	3245	484149	99.334214	2266
40	BorderFAF2	float64	4172	483222	99.144019	129
39	BorderFAF1	float64	4172	483222	99.144019	127
26	Truck	object	4365	483029	99.104421	3
38	BorderState2	object	5863	481531	98.797072	46
37	BorderState1	object	5983	481411	98.772451	47
12	Rte_Qualifier	object	7361	480033	98.489723	9
31	Toll_Name	object	7992	479402	98.360259	379
30	Toll_Type	float64	8012	479382	98.356155	3
44	AdjustSpeed	float64	38858	448536	92.027395	71
45	AdjustReason	object	39206	448188	91.955995	14
25	NHFN	float64	86859	400535	82.178894	4
35	HPMS_Begin_Point	float64	99440	387954	79.597615	86891
36	HPMS_End_Point	float64	99440	387954	79.597615	87840
34	HPMS_USA_RouteID	object	99463	387931	79.592896	10253
24	STRAHNET	float64	101707	385687	79.132488	5
28	BA_Lanes	float64	153802	333592	68.444010	10
9	Sign_Rte	object	236561	250833	51.464113	5372
10	Rte_Type	object	236582	250812	51.459805	9
11	Rte_Number	object	236603	250791	51.455496	1472
23	NHS	float64	322026	165368	33.929018	9
22	Facility_Type	float64	393114	94280	19.343693	7
8	Road_Name	object	448694	38700	7.940188	24320
27	AB_Lanes	float64	463206	24188	4.962720	17
21	F_Class	float64	465623	21771	4.466817	8
18	Urban_Code	float64	481840	5554	1.139530	505

	column	dtype	non_null	missing	missing_pct	unique_values
16	County_Name	object	481840	5554	1.139530	1836
17	CTFIPS	float64	481840	5554	1.139530	3138
51	BA_FreeFlowTime	float64	483542	3852	0.790326	483236
20	Status	float64	483598	3796	0.778836	4
48	AB_CombinedSpeed	float64	483875	3519	0.722003	82
47	BA_FinalSpeed	float64	483875	3519	0.722003	474
46	AB_FinalSpeed	float64	483875	3519	0.722003	160
50	AB_FreeFlowTime	float64	483875	3519	0.722003	483594
49	BA_CombinedSpeed	float64	483875	3519	0.722003	81
15	STFIPS	float64	485636	1758	0.360694	51
19	FAFZONE	float64	485636	1758	0.360694	132
29	Speed_Limit	float64	487162	232	0.047600	18
7	Class_Description	object	487390	4	0.000821	31
6	Class	float64	487390	4	0.000821	16
4	DATA1	int64	487394	0	0.000000	487394
14	STATE	object	487394	0	0.000000	58
13	Country	object	487394	0	0.000000	2
5	VERSION	object	487394	0	0.000000	1
0	OBJECTID	int64	487394	0	0.000000	487394
1	ID	int64	487394	0	0.000000	487394
2	LENGTH	float64	487394	0	0.000000	485276
3	DIR	int64	487394	0	0.000000	3
52	Shape_Length	float64	487394	0	0.000000	487335

	OBJECTID	ID	LENGTH	DIR	DATA1	Class	
count	487394.000000	4.873940e+05	487394.000000	487394.000000	4.873940e+05	487390.000000	48563
mean	243697.500000	9.748118e+05	1.206835	0.653264	8.932146e+05	15.903156	2
std	140698.672888	5.627960e+05	3.584868	0.477614	3.461475e+05	5.162112	1
min	1.000000	1.000000e+00	0.000000	-1.000000	2.000000e+00	11.000000	
25%	121849.250000	4.874472e+05	0.126454	0.000000	6.956222e+05	12.000000	1
50%	243697.500000	9.748635e+05	0.295451	1.000000	9.600420e+05	14.000000	3
75%	365545.750000	1.462280e+06	0.781605	1.000000	1.178870e+06	22.000000	4
max	487394.000000	1.949504e+06	739.858582	1.000000	1.359962e+06	50.000000	5



Top 3 columns with outliers (Magnitude X IQR): TRUCKTOLL , AB_FreeFlowTime , BA_FreeFlowTime

These fields require business-context validation because some high values are associated with ferry or long-distance links and may be valid rather than erroneous.

--- STATISTICAL OUTLIER DETECTION (IQR METHOD) ---

	Column	Min	Max	Outliers Count	Outliers (%)	Outlier Magnitude	Outlier Magnitude (x IQR)
0	LENGTH	0.000000e+00	739.858582	73126	15.00	738.09	1126.60
29	Shape__Length	9.996506e-07	17.175784	72846	14.95	17.15	1579.87
27	AB_FreeFlowTime	0.000000e+00	4439.151489	71892	14.75	4436.76	5054.70
28	BA_FreeFlowTime	-6.426970e+00	4439.151489	71836	14.74	4436.74	5017.68
10	AB_Lanes	0.000000e+00	17.000000	45954	9.43	13.50	13.50
15	HPMS_Begin_Point	-6.870000e+00	878.780650	12701	2.61	706.81	10.55
16	HPMS_End_Point	-6.528595e+00	879.427677	12696	2.60	705.88	10.49
2	Class	1.100000e+01	50.000000	3986	0.82	13.00	1.30
11	BA_Lanes	0.000000e+00	12.000000	3826	0.78	8.50	8.50
5	F_Class	1.000000e+00	10.000000	1454	0.30	4.00	2.00
19	TRUCKTOLL	2.366000e-03	6789.000000	236	0.05	6781.56	2314.86
22	AdjustSpeed	-1.220000e+01	-1.391230	192	0.04	2.26	0.82
13	Toll_Type	1.000000e+00	3.000000	196	0.04	2.00	NaN
4	Status	1.000000e+00	5.000000	28	0.01	4.00	NaN
24	BA_FinalSpeed	-6.750000e+00	83.000000	8	0.00	1.00	0.04

3. Data Quality Findings and Identified Issues

Schema is not consistend and contains many missing rows. Most of them can be grouped into data silos.

Finding 1: Missing Values and Data Silos

	Missing Rows	Percentage (%)
BorderLink	487226	99.965531
AddedBorderTime	487226	99.965531
Toll_Link_Name	486259	99.767129
Toll_Link	486151	99.744970
TRUCKTOLL	484149	99.334214
BorderFAF2	483222	99.144019
BorderFAF1	483222	99.144019
Truck	483029	99.104421
BorderState2	481531	98.797072
BorderState1	481411	98.772451
Rte_Qualifier	480033	98.489723
Toll_Name	479402	98.360259
Toll_Type	479382	98.356155
AdjustSpeed	448536	92.027395
AdjustReason	448188	91.955995
NHFN	400535	82.178894
HPMS_Begin_Point	387954	79.597615
HPMS_End_Point	387954	79.597615
HPMS_USA_RouteID	387931	79.592896
STRAHNET	385687	79.132488
BA_Lanes	333592	68.444010
Sign_Rte	250833	51.464113
Rte_Type	250812	51.459805
Rte_Number	250791	51.455496
NHS	165368	33.929018
Facility_Type	94280	19.343693
Road_Name	38700	7.940188
AB_Lanes	24188	4.962720
F_Class	21771	4.466817
Urban_Code	5554	1.139530
County_Name	5554	1.139530
CTFIPS	5554	1.139530
BA_FreeFlowTime	3852	0.790326
Status	3796	0.778836
AB_CombinedSpeed	3519	0.722003
BA_FinalSpeed	3519	0.722003
AB_FinalSpeed	3519	0.722003
AB_FreeFlowTime	3519	0.722003
BA_CombinedSpeed	3519	0.722003
STFIPS	1758	0.360694
FAFZONE	1758	0.360694
Speed_Limit	232	0.047600
Class_Description	4	0.000821
Class	4	0.000821
DATA1	0	0.000000
STATE	0	0.000000
Country	0	0.000000
VERSION	0	0.000000
OBJECTID	0	0.000000
ID	0	0.000000
LENGTH	0	0.000000
DIR	0	0.000000
Shape__Length	0	0.000000

--- FULL AUDIT OF STRUCTURE AND DATA SILOS ---

Total columns in the dataset: 53

 COLUMN DISTRIBUTION BY DATA SILO CLUSTERS (FOR REPORTING):

 Cluster: 0% (Perfect/Core Fields)

Columns (9 total): DATA1, STATE, Country, VERSION, OBJECTID, ID, LENGTH, DIR, Shape__Length

 Cluster: 0% - 8% (Minor Incompleteness)

Columns (18 total): Road_Name, AB_Lanes, F_Class, Urban_Code, County_Name, CTFIPS, BA_FreeFlowTime, Status, AB_CombinedSpeed, BA_FinalSpeed, AB_FinalSpeed, AB_FreeFlowTime, BA_CombinedSpeed, STFIPS, FAFZONE, Speed_Limit, Class_Description, Class

 Cluster: 51% - 52% (Specific Infrastructure Silo)

Columns (4 total): Sign_Rte, Rte_Type, Rte_Number, NHS

 Cluster: 68% (Niche Network Attributes)

Columns (1 total): BA_Lanes

 Cluster: 79% - 82% (High Sparsity / Federal Sub-networks)

Columns (5 total): NHFN, HPMS_Begin_Point, HPMS_End_Point, HPMS_USA_RouteID, STRAHNET

 Cluster: 8% - 33% (Regional/State Specific Data)

Columns (1 total): Facility_Type

 Cluster: 91% - 92% (Operational Adjustments Silo)

Columns (2 total): AdjustSpeed, AdjustReason

 Cluster: 98% - 100% (Critical Sparsity / Edge Cases / Border Only)

Columns (13 total): BorderLink, AddedBorderTime, Toll_Link_Name, Toll_Link, TRUCKTOLL, BorderFAF2, BorderFAF1, Truck, BorderState2, BorderState1, Rte_Qualifier, Toll_Name, Toll_Type

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--- SCHEMA DRIFT ANALYSIS BY VERSION COLUMN ---

Detected versions/reporting periods in the 2026 data: ['V2021.05']

Analysis of version subset V2021.05 (Rows: 487394):

BorderLink	99.965531
AddedBorderTime	99.965531
Toll_Link_Name	99.767129
Toll_Link	99.744970
TRUCKTOLL	99.334214
dtype:	float64

=====

--- MISSINGNESS CORRELATION (Identifying Systemic Data Silos) ---

Found 24 column pairs that become empty AT THE SAME TIME (evidence of strict data silos):

- Class <--> Class_Description
- Sign_Rte <--> Rte_Type
- Sign_Rte <--> Rte_Number
- Rte_Type <--> Rte_Number
- STFIPS <--> FAFZONE

- County_Name <---> CTFIPS
- County_Name <---> Urban_Code
- CTFIPS <---> Urban_Code
- HPMS_USA_RouteID <---> HPMS_Begin_Point
- HPMS_USA_RouteID <---> HPMS_End_Point
- HPMS_Begin_Point <---> HPMS_End_Point
- BorderFAF1 <---> BorderFAF2
- BorderLink <---> AddedBorderTime
- AdjustSpeed <---> AdjustReason
- AB_FinalSpeed <---> BA_FinalSpeed

The missingness correlation analysis found **24 column pairs** that become empty at the same time, which is evidence of structured data silos. Examples include:

- Sign_Rte <--> Rte_Type
- Sign_Rte <--> Rte_Number
- Rte_Type <--> Rte_Number
- STFIPS <--> FAFZONE
- County_Name <--> CTFIPS
- BorderLink <--> AddedBorderTime

These are data silos, not random groups: some fields are populated only for certain network types, geographies, or enrichment sources.

Finding 2: Inaccurate or Inconsistent Data

--- LOGICAL INCONSISTENCIES ---

- ⚠ Negative values found in strictly positive fields:
 - Column 'DIR': 391 rows with negative values.
 - Column 'HPMS_Begin_Point': 25 rows with negative values.
 - Column 'HPMS_End_Point': 22 rows with negative values.
 - Column 'AdjustSpeed': 38858 rows with negative values.
 - Column 'BA_FinalSpeed': 26 rows with negative values.
 - Column 'BA_FreeFlowTime': 26 rows with negative values.
 - Rows with unrealistic Speed_Limit (≤ 0 or > 90): 0
 - Rows with zero or negative LENGTH: 2

Finding 3: Inconsistent Class_Description Values

Class_Description appears to contain manual or non-standardized text variants.

	Class	Class_Description	rows
4	14.0	Arterial or Major Collector	200129
6	14.0	Frontage/Service Road	3
3	14.0	Arterial or Major Collec	1
5	14.0	Artial or Major Collector	1
13	16.0	Frontage/Service Road	11226
15	16.0	Frontage/Servise Road	13
9	16.0	Collector/Distributor Lane	12
11	16.0	Frontage/Servce Road	12
8	16.0	Arterial or Major Collector	6
12	16.0	Frontage/Service Rd	2
14	16.0	Frontage/Service Roaed	2
17	16.0	Froontage/Service Road	2
10	16.0	Frontage /Service Road	1
16	16.0	Frontasge/Service Road	1
18	16.0	Washington Ave	1
22	19.0	Facility Access/Circulator Road	1804
21	19.0	Arterial or Major Collector	1
23	19.0	Facility Accss/Circulator Road	1
24	19.0	Facilty Access/Circulator Road	1
28	23.0	Collector/Distributor Lane	4578
27	23.0	Collector/Distirbutor Lane	12
30	33.0	Express Lane (Truck Route)	12
31	33.0	Express Lane (Truck Toll)	3
29	33.0	Express Lane (Truck Only)	2

Finding 4: Non-USA Geography Fields Have 100% Missing Values for Some Attributes

fields are missing: STFIPS , County_Name , CTFIPS , Urban_Code , and FAFZONE .

	column	missing_pct
0	Country	0.0
1	STATE	0.0
2	STFIPS	100.0
3	County_Name	100.0
4	CTFIPS	100.0
5	Urban_Code	100.0
6	FAFZONE	100.0

Finding 5: Directional Fields Are Structurally Difficult to Interpret

There are inconsistencies in the **AB_Lanes** and the **BA_Lanes** records in assumption that **DIR** have correspondence with them.

For instance, there are cases when **AB_Lanes** and **BA_Lanes** are empty at the same time. Other combinations of missing values tables are shown below.

AB_Lanes_missing		False		True	
BA_Lanes_missing	False	True	False	True	
DIR					
	-1	110	234	0	47
	0	152972	178	29	15036
	1	684	309028	7	9069

Assuming that **DIR** =0 means two-way movement and **DIR** =1 means one-way movement, combinations in red look like an ambiguity and require verification.

AB_Lanes_missing	False		True	
BA_Lanes_missing	False	True	False	True
DIR				
-1	110	234	0	47
0	152972	178	29	15036
1	684	309028	7	9069

Finding 6: Completeness Drift Across Reporting or Load Periods

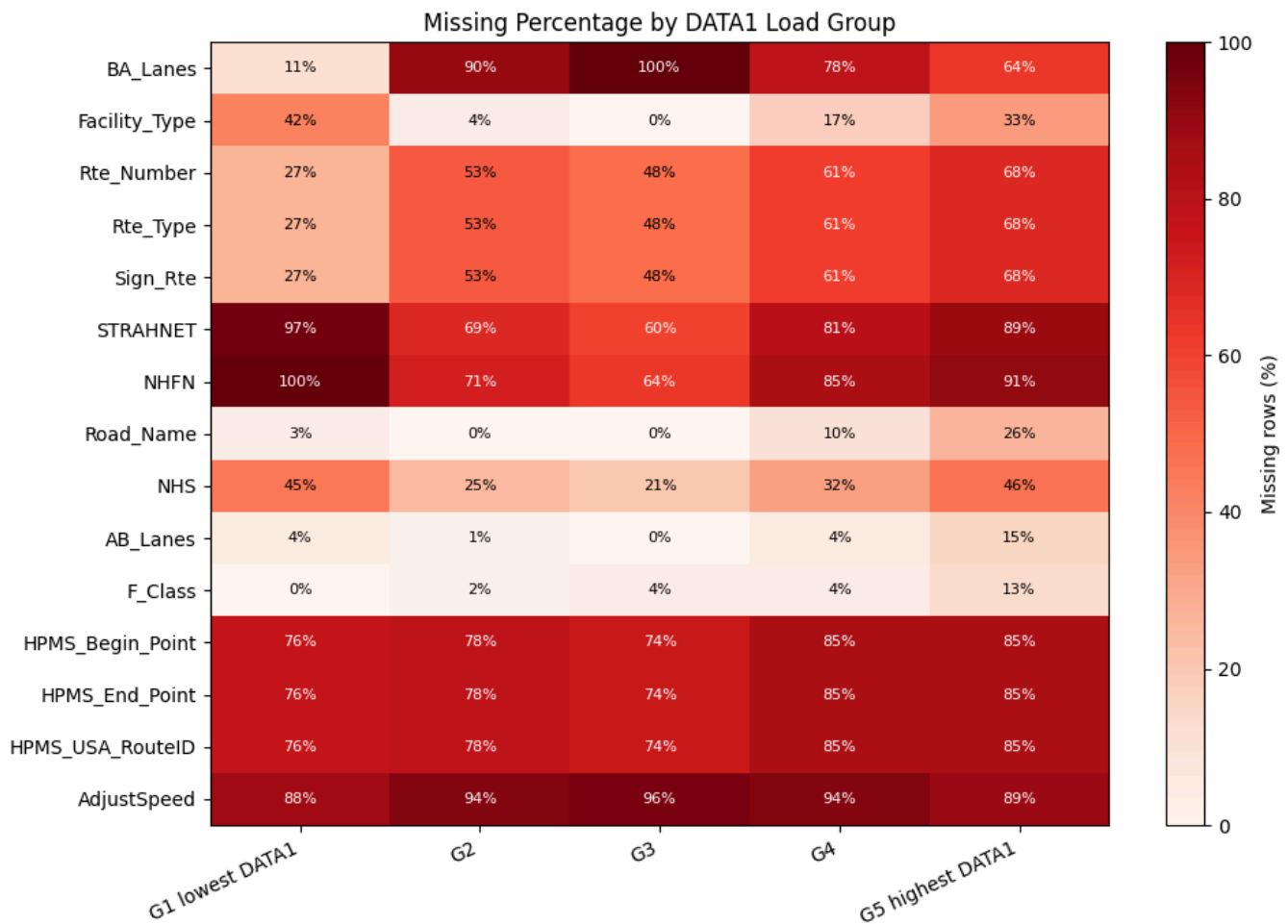
Assuming `DATA1` it is a dummy timestamp column, we divide our dataframe into 5 equal subsets and compare their % of missing data. The columns and schema are the same, however, missing data drift was detected.

Subsets

	_DATA1_Load_Group	Rows	Min_DATA1	Max_DATA1	Unique_States	Unique_Classes
0	G1 lowest DATA1	97479	2	613485	48	9
1	G2	97479	613495	852437	32	14
2	G3	97478	852439	1058601	36	13
3	G4	97479	1058602	1221797	48	12
4	G5 highest DATA1	97479	1221798	1359962	57	16

All the columns in the heatmap below have different missing values amount across periods. The most outstanding of them are:

- `BA_Lanes` missing rate changes from roughly 11% to almost 100%.
- `Facility_Type` missing rate changes from near 0% to above 40%.
- `Road_Name` missing rate changes from near 0% to about 26%.



Finding 7: Temporal Mismatch: task asks for a 2026 evaluation, but all records at the task's link have `VERSION = V2021.05` with 2022 publication date.

Finding 8: Absence of geometry columns (such as longitude and latitude)

4. Risk Assessment Related to Data Reliability and Usability

Findings:

- Missing Values and Data Silos:** High missingness (up to 50%) in almost 50% of the columns.
- Inaccurate or Inconsistent Data:** negative speed, `DIR`, unrealistic speed limits.
- Inconsistent `Class_Description` Values
- Non-USA Geography Fields Have 100% Missing Values** for `STFIPS`, `County_Name`, `CTFIPS`, `Urban_Code`, and `FAFZONE`
- Directional Fields Are Structurally Difficult to Interpret:** `AB_Lanes`, `BA_Lanes` in relation to `DIR` such as other `AB_`, `BA_` columns.

6. **Completeness Drift Across Reporting Periods:** data missings drifts depend on DATA1 (in assumption this is proxy timestamp load)
7. **Temporal Mismatch:** task asks for a 2026 evaluation, but all records at the task's link have `VERSION = V2021.05` with 2022 publication date.
8. **Absence of geometry columns** (such as longitude and latitude)

Production Risk: Medium-to-high

5. Recommendations for Data Cleansing, Governance, and Validation

Corresponding Recommendations:

1. **Implement automated data-quality checks.**

Add automated ingestion checks for uniqueness, missingness thresholds, code-domain validation, numeric range validation, schema drift, and batch-level completeness drift.

2. **Standardize categorical fields.**

Standardize `Class_Description` using an official code-value lookup table. The same for `DIR`, `Status`, `Facility_Type`, `NHS`, `NHFN`, `STRAHNET`, `Toll_Type`, and `Truck`.

3. **Define clear missing-value rules.**

Non-USA fields which are not applicable should include 'Not Applicable' to avoid perceiving them as missing data. Missing values should distinguish between unknown, not applicable, and not collected. Treat non-USA rows and cross-border records as a separate integration case.

4. **Validate directional fields.**

such as `AB_Lanes`, `BA_Lanes`, `AB_FinalSpeed`, `BA_FinalSpeed`, `AB_FreeFlowTime`, and `BA_FreeFlowTime` against official rules for one-way and two-way links.

5. **Resolve the reporting-date mismatch.**

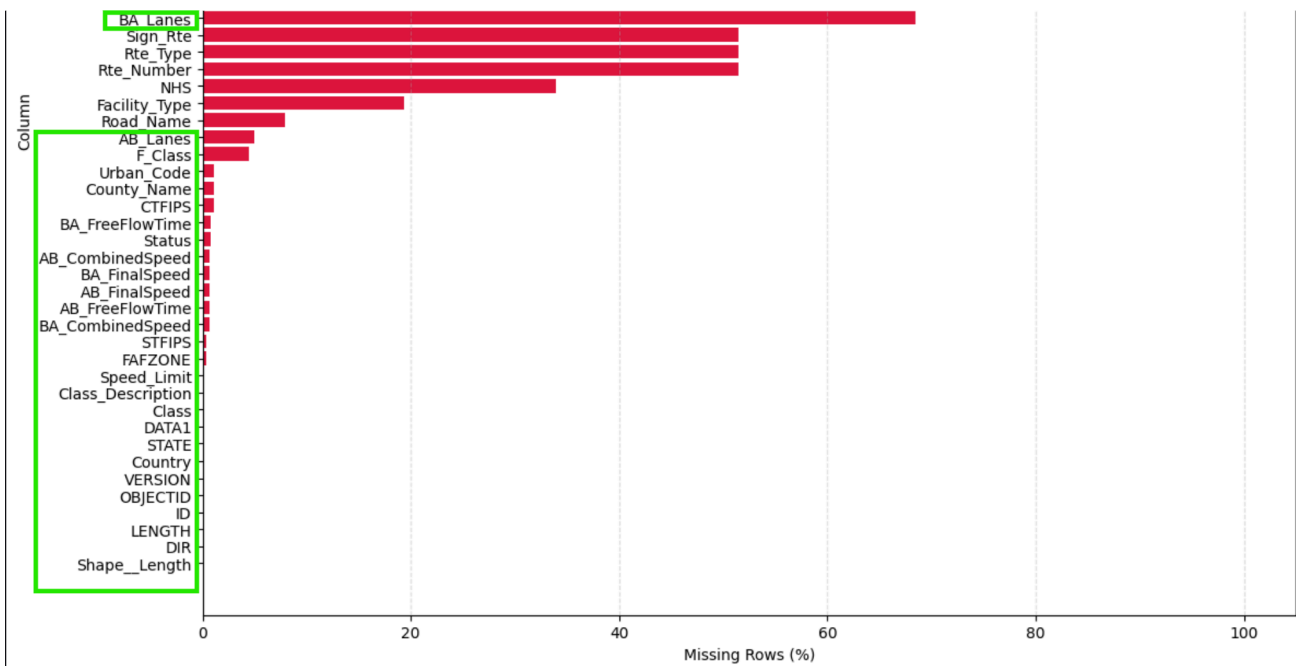
Resolve the 2026 versus 2022 date mismatch: The dataset should include a verified `data_year` field

6. **Use authoritative geometry.**

Use geometry from the authoritative source geodatabase for geospatial validation and map-based analytics

6. Conclusion on Production-Grade Analytics Readiness

The dataset is suitable only for **limited exploratory analysis and approximate transportation network modeling** if the columns highlighted in green in the snapshot below are sufficient for the intended purpose.



However, the dataset is **not reliable as-is for a production-grade analytics platform** without additional validation, documentation, and governance.

The main reasons are the lack of confirmed 2026 temporal metadata, missingness in important fields, inconsistent categorical labels, ambiguous directional fields, and the absence of full geometry.

These issues can affect routing, travel-time estimation, cost modeling, reporting, and integration with external reference datasets.

Final recommendation: **reject the current dataset version for production use until temporal validity, schema documentation, and data-quality rules are confirmed.** After cleansing and governance, it could be used in production-grade analytics platform.